

# Development of a Meta-Question Enhanced Retrieval-Augmented Generation Model and Its Application in Dermatology

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**Abstract**—RAG (Retrieval Augmented Generation) provides a solution for enhancing domain-specific knowledge in large language models by interacting with external information. RAG functions similarly to a search engine, retrieving the most relevant knowledge or related conversation history to the user's query. It then combines this with the original query to create a detailed prompt, guiding the model to generate accurate outputs. However, traditional RAG methods match short queries with long texts in the knowledge base, leading to mismatched results despite apparent similarities due to length discrepancies. To address this issue, we propose the MQ-RAG (Meta Questions Retrieval Augmented Generation) method, a question-answer pair-based approach built upon RAG. The key difference in MQ-RAG lies in the retrieval and recall content: it uses a database of Meta Questions, which are pairs of questions and corresponding answers. When using MQ-RAG, queries are matched with Meta Questions to retrieve the most relevant answers, ensuring high compatibility between the query and the retrieved content. For evaluation, 100 multiple-choice questions from the Dermatology and Venereology Medical Licensing Examination database were selected. Results showed that, using Qwen2-7b and Qwen2-72b as base models, MQ-RAG improved accuracy by 17.14% and 2.27%, respectively, compared to traditional RAG. Our MQ-RAG method demonstrated higher accuracy across both large and small parameter models, with particularly significant improvements in smaller models.

**Keywords**—Large Language Model, Meta Questions, Medical Dermatology Department

## I. INTRODUCTION

As large language models gain popularity, increasing efforts are being directed toward enhancing these models for specific vertical domains[1], such as the medical field[2]. Typically,

domain-specific adjustments to large language models are accomplished using two methods: Retrieval-Augmented Generation (RAG) and fine-tuning. These methods aim to address the inherent knowledge limitations of large language models in specialized areas[3].

In a RAG system, documents are segmented into multiple text blocks before vector embedding. Without considering the constraints of model input length and cost, the goal is to minimize noise in the embedded content while maintaining semantic coherence, thereby more effectively locating the most relevant document sections for user queries. If the blocks are too large, they may contain excessive irrelevant information[4], reducing retrieval accuracy. Conversely, if the blocks are too small, essential context might be lost, leading to responses that lack coherence or depth. An appropriate document segmentation strategy in a RAG system seeks to balance information completeness and relevance. Ideally, text blocks should be meaningful to humans without requiring surrounding context, ensuring they are also meaningful to the language model. Generally, fixed-size blocks are chosen, as this is the simplest and most straightforward method. RAG typically sets a fixed word count for the blocks and decides whether to repeat content between blocks, often maintaining some overlap to ensure semantic context is preserved.

User-submitted queries are generally brief, while the document chunks used in the RAG method are relatively long. When performing retrieval matching between short queries and long texts, the discrepancy in text length often results in the RAG method returning irrelevant or unrelated results, thereby diminishing the knowledge base's effectiveness in responding to user queries. Aligning short queries with RAG documents is

therefore crucial; however, directly aligning a RAG document with a user query is challenging.

To address the issue of matching short queries with long texts, we proposed the MQ-RAG method. This method involves generating Meta Questions—essentially question-answer pairs—from the document content. During the similarity matching process with a user's query, our generated questions are matched against the query, and the retrieved content consists of the corresponding answers. Unlike traditional RAG, MQ-RAG does not rely on text-based searches but instead matches the user's query with our Meta Questions to ensure highly relevant retrieval[5].

This paper makes the following contributions: (a) It proposes the MQ-RAG method to enhance large language models in specific domains by using Meta Questions to align with user queries. (b) In the professional dermatology field, 100 test questions were selected from the dermatology and venereology chief physician examination question bank for evaluation. The models using the MQ-RAG method, Qwen-7b-MQ-RAG and Qwen-72b-MQ-RAG, were compared with GPT-4o, Qwen-7b, Qwen-72b, Qwen-7b-RAG, and Qwen-72b-RAG. The results demonstrate that models utilizing MQ-RAG showed significant performance improvement..

## II. RELATED WORK

Significant progress has been made by large language models like GPT-4 across various natural language processing tasks, demonstrating strong performance on general knowledge points. However, when applied to specific domains, their effectiveness is often limited. This limitation arises primarily due to the lack of domain-specific details and knowledge in the datasets used during the pre-training phase, leading to insufficient domain-specific understanding. Retraining these models from scratch is a labor-intensive task with limited generalization capabilities. As a result, optimizing these models for specific domains has become a critical focus of research.

Existing methods for fine-tuning large language models for specific domains can be categorized into two main approaches: (1) fine-tuning and (2) Retrieval-Augmented Generation (RAG) [6]. Fine-tuning involves adjusting all or part of the parameters of a pre-trained model using a small dataset to achieve better performance in a specific domain. While effective, fine-tuning presents several challenges. First, as generative models rely on their internal knowledge (weights), they are prone to producing hallucinations [7], which is unacceptable in scenarios requiring high accuracy and understanding. Users often cannot discern whether the model's responses are accurate or nonsensical. Second, in real-world applications, data is generated rapidly and concepts evolve quickly, such as policy interpretations or metric adjustments. Fine-tuning is a complex task, involving data preparation, computational resources, effectiveness of tuning, and training time, making it impractical to continuously update the model with new data [8].

An alternative solution, RAG, offers a promising way for generative models to interact with the external world. RAG functions similarly to a search engine, retrieving the most relevant knowledge or dialogue history in response to user queries and combining this with the original query to create an

informative prompt, guiding the model to generate accurate outputs. Essentially, RAG applies the principle of In-Context Learning [9].

However, in practical application, the addition of RAG does not always yield the expected results. It has been observed that the mismatch between the short user queries and the lengthy RAG document contents leads to poor alignment [10], causing relevant recalls to retrieve unrelated content. To address this issue, we propose a new method: MQ-RAG. Building on our previous research in dermatology such as syphilis [11], we applied the proposed MQ-RAG model in this field as a case study.

## III. METHODOLOGY

This section outlines the specific steps of our work. The overall framework of our MQ-RAG method is illustrated in Figure 1. Initially, we selected medical guideline documents related to dermatology and logically segmented them into well-structured paragraphs. Each paragraph was transformed into a coherent long text. From these segmented texts, high-quality dermatology question-and-answer pairs were generated, which formed the content of our documents. These Q&A pairs were then encoded, with the encoded content stored in the Faiss vector database for later use. When a user submits a query, it is matched against the encoded Meta Questions in our database to identify the question most similar to the user's query. The corresponding answer to that question is retrieved as the knowledge content for our large model. Thus, the input to the large language model includes both the user's query and the relevant answer content. This method of incorporating an additional knowledge base enhances the model's output.

Next, we will detail the MQ-RAG method. We first obtained dermatology-related medical guidelines and used Python's PyMuPDF to logically segment the guideline documents, saving the segmented content into corresponding txt files. We then prepared the Meta Questions by generating question-and-answer pairs. The txt files were input into the GPT-4o model using the following prompt: [If you were a medical document extractor, I would now like you to help me extract all the text completely and convert it into a question-and-answer format, with one question corresponding to one answer, ensuring that no information in the txt text is omitted. I want as many questions as possible, even if the meanings are similar but described differently, there can be different forms of questions and answers. The more detailed and lengthy the answers, the better. The answers need to be reasonably expanded and polished based on the relevant content in the text to ensure sufficient word count. Ensure that the question and answer pairs cover all the content of the entire document. Multiple generations can be performed. Please process from the beginning of the txt to the end. No need to display titles, just the question-and-answer format.]

The generated question-and-answer pairs were saved into a JSON file. The content was then encoded and stored in a Faiss vector database.

We compared the performance of a regular large language model with the MQ-RAG enhanced model. The testing process was as follows:

- **Training:**  $Q \rightarrow A$

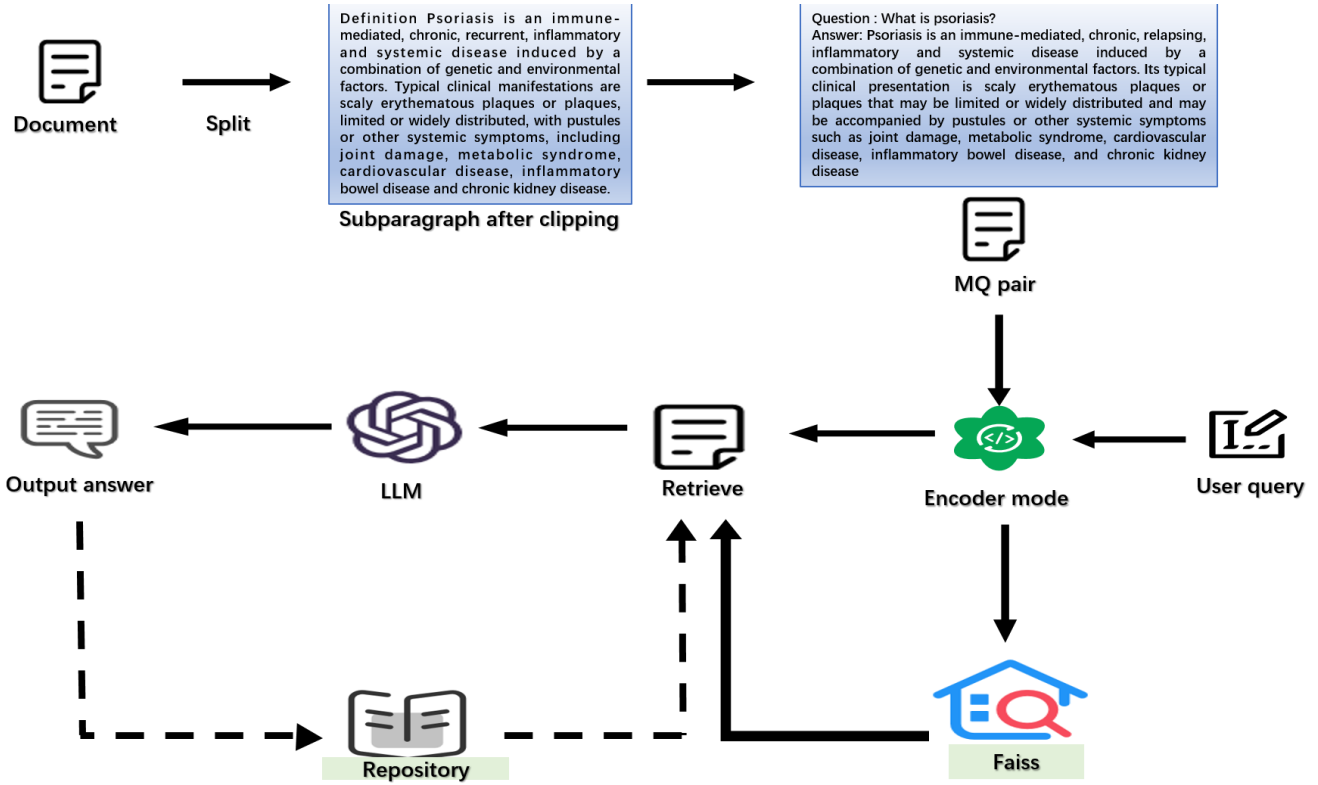


Fig. 1. MQ-RAG Framework Process. Encoder model refers to the encoding of MQ pair in the text section. The llm input actually includes the user's input, the answer document corresponding to the most relevant question of the user's input question.

- **Inference 0:**  $Q \rightarrow A$
- **MQ-RAG inference:**  $Q + D \rightarrow A$

When a user query is received, it is searched within our retriever. The retriever is designed to find relevant documents from the knowledge base. Given a query  $q$  and a set of question-answer pairs  $D$  in the knowledge base, the retriever's goal is to find the question-answer pairs in  $D$  most relevant to  $q$ .

$$P(d|q) = \frac{\exp(\text{sim}(q, d))}{\sum_{\{d' \in D\}} \exp(\text{sim}(q, d'))} \quad (1)$$

Here,  $\text{sim}(q, d)$  represents the similarity between the query and the document. Common similarity measures include cosine similarity or dot product.

After completing the retrieval process, the next step involves generating answers using the model. Given a query  $q$  and the retrieved document  $d$ , the model aims to generate an answer or text. The objective of the generator model is to compute the conditional probability  $P(a|q, d)$ , where  $a$  represents the generated answer.

$$P(a|q, d) = \prod_{t=1}^T P(a_t|q, d, a_{\{<t\}}) \quad (2)$$

Here,  $a_{\{<t\}}$  represents the word generated at time step  $t$ , and  $a < t$  represents all the words generated before time step  $t$ .

After retrieval and generation are completed, the results from the retriever and generator must be combined to compute the final probability of the generated answer  $a$  given the query  $q$ .

$$P(a|q) = \sum_{d \in D} P(a|q, d)P(d|q) \quad (3)$$

This involves calculating the weighted sum of the generation probabilities  $P(a|q, d)$  and the retrieval probabilities  $P(d|q)$  for all possible retrieved documents  $d$ .

By employing these methodological steps, the MQ-RAG model efficiently retrieves relevant question-answer pairs from large-scale knowledge bases and generates high-quality answers using the generative model.

#### IV. EXPERIMENT

In this section, we will present the specific details of our experimental process, along with the testing results of the MQ-RAG method and other models within the Dermatology and Venereology Physician Examination question bank.

First, we encoded the generated question-answer pairs using the `nlp_corom_sentence-embedding_chinese-base-medical` encoder [12], which is specialized for the Chinese medical domain. The encoded data was then stored in Faiss [13], a library designed for large-scale vector retrieval and similarity search, enabling rapid identification of the most similar vectors within massive datasets. We then employed `bge-reranker-large` as our reranked model [14]. During the reranked retrieval process [15], we retrieved the question-answer pairs and used the

corresponding answer content as external knowledge when processing user queries. This marks the completion of the MQ-RAG implementation within the large language model.

Next, we detail the parameter settings. During the reranked retrieval, the parameters were set to `top_choice = 5` and `score_choice = 3.0` to ensure the high quality of the retrieved question-answer pairs. The prompt used was as follows: ' {query}. The above is a dermatology-related multiple-choice question and its options. Please provide the best choice for each question. Each question must have one selected option. Document Q&A content: "{source} ". If the question is relevant to the document Q&A, please select the correct option based on the document Q&A ordered by relevance. If the question is unrelated to the document Q&A, please provide the answer based on common medical knowledge of dermatological diseases.' For the final testing, the large language model parameters were set as follows: `max_new_tokens = 1920`, `top_p = 0.8`, `top_k = 20`, and `temperature = 0.9`.

We then describe the detailed prompt settings and what is returned when testing a large language model using the MQ-RAG approach[16]. The detailed test case is shown in Figure 2

```
{'role': 'system', 'content': 'You are a medical dermatologist and your task is to answer a medical multiple choice question posed by USER, the answer needs to be specific and medically based.'}

-----

Input_query: f{query}. These are the multiple choice questions in medical dermatology along with the option content, please output the best option that should be selected for each question, you must output one option for each question. Documentation quiz pair content: '{source}'. If the question is relevant to the document quiz pair, you can select the correct option based on the following document quiz pairs provided in order of relevance. \n If the question has no relevance to the document quiz pair, please provide your answer based on medical knowledge of common dermatological disorders.'
```

```
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User query: {"question": "In the treatment of cardiovascular syphilis, the drug to be administered 1 port before the penicillin injection to avoid a gihai reaction is () O", "A": "Prednisone", "B": "Paracetamol", "C": "Chlorpromazine", "D": "Probenecid", "E": "Epinephrine"}
```

```
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Output: In cardiovascular syphilis therapy, the drug to be administered 1 port before penicillin injection to avoid gihai reaction is A. Prednisone.
```

Fig. 2. MQ-RAG prompt operation, query represents the user's query, and the source content is the content of the answer corresponding to the question.

We conducted three tests on each model, calculating the accuracy and standard deviation for each run. The specific results are shown in Table 1. The analysis of the test results reveals key insights into the performance and stability of various models. The Qwen2-72b and Qwen2-7b models, without RAG

or MQ-RAG enhancements, displayed greater performance fluctuations[17], indicating less consistent results when handling the same tasks. The Qwen2-72b-MQ-RAG model emerged as the most stable, evidenced by its lowest standard deviation (0.014) across multiple tests, indicating consistent performance. In contrast, the Qwen2-7b-RAG model showed the highest variability, with a standard deviation of 0.065, suggesting less stability in its results. When considering overall performance, the Qwen2-72b-MQ-RAG model again outperformed others, achieving the highest average score of approximately 0.899, highlighting its superior efficacy. The average accuracy across all tests is summarized in Figure 3.

TABLE I. MODEL TESTING ACCURACY

| Models           | Test details |        |       | Standard deviation |
|------------------|--------------|--------|-------|--------------------|
|                  | First        | Second | Third |                    |
| GPT4-o           | 0.758        | 0.788  | 0.818 | 0.025              |
| Qwen2-7b         | 0.697        | 0.758  | 0.818 | 0.049              |
| Qwen2-72b        | 0.727        | 0.848  | 0.758 | 0.051              |
| Qwen2-7b-RAG     | 0.606        | 0.727  | 0.758 | 0.065              |
| Qwen2-72b-RAG    | 0.848        | 0.939  | 0.848 | 0.043              |
| Qwen2-7b-MQ-RAG  | 0.788        | 0.818  | 0.848 | 0.025              |
| Qwen2-72b-MQ-RAG | 0.909        | 0.909  | 0.879 | 0.014              |

During testing, we observed some interesting phenomena:

(a) Regardless of whether the model had a smaller or larger parameter size, the use of the MQ-RAG method resulted in significant improvements compared to the original model. When compared to models using RAG, our MQ-RAG method also demonstrated enhanced performance, indicating its effectiveness in domain-specific enhancement;

(b) The enhancement effects varied depending on the model's parameter size. For smaller models, MQ-RAG provided a substantial improvement, while for larger models, the

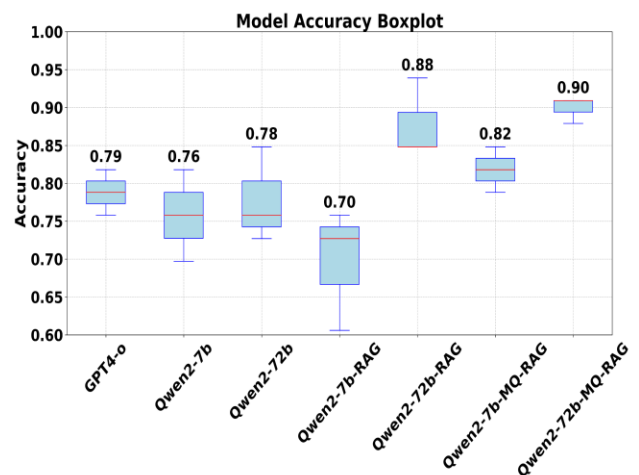


Fig. 3. Test results.

enhancement was more incremental. This is likely because smaller models tend to be less capable in terms of logic and basic abilities, making our method more impactful. Interestingly, the use of RAG in smaller models sometimes resulted in worse performance, likely due to the lower cognitive capabilities of these models. When provided with overly lengthy documents by RAG, smaller models struggled to process the information, leading to diminished performance[18];

(c) It is evident that the standard deviation across the three tests conducted with the MQ-RAG method is significantly lower compared to other models, indicating that our approach effectively reduces model uncertainty and enhances stability.

## V. CONCLUSION

This paper proposes the MQ-RAG method, which matches user queries with Meta Questions to better align retrieval content with the query and extract the corresponding answer as knowledge. When users input a question, the relevant answer is retrieved and used to respond. We conducted performance tests using 100 questions from the Dermatology and Venereology Medical Licensing Examination database. The results indicate that the MQ-RAG method significantly enhances the model's capabilities in the relevant field. MQ-RAG represents a major advancement in content retrieval for large language models, offering a valuable tool for enhancing specific domains across various industries. In future work, we may start with more aspects[19], such as fine-tuning the model or improving the performance of the model by introducing additional modules for the attention mechanism [20].

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